

2019 Ph H2 Q15

Section: Our Dynamic Universe

Topic: Oscillations of a physical pendulum

A 1.00 m wooden rod is suspended on a pin through a hole 50 mm from one end. When displaced and released, it oscillates with period T . (a) Method to obtain accurate T with stopwatch. (b) Graph of T vs h , use to determine values of h for given T and estimate minimum T . (c) Use relationship $T^2 = (4\pi^2/g)(h^2 + C)$ to calculate constant C when $h=0.30$ m.

Worked solution

(a) To obtain an accurate value of T using a stopwatch:

- Start timing when the rod passes the central position.
- Measure the total time for, say, 20 oscillations.
- Divide by 20 to obtain the average period.

This reduces the effect of human reaction time.

Answer: Measure many oscillations, divide to find mean period.

(b)(i) A graph of T against h should be plotted (points given in table). T decreases with h , then increases again, forming a U-shaped curve.

(b)(ii) From the graph, two values of h give $T = 1.57$ s. By inspection, $h \approx 0.41$ m and $h \approx 0.20$ m.

Answer: 0.41 m and 0.20 m

(b)(iii)(A) From the graph, the minimum period is about 1.53 s.

Answer: $T_{\min} \approx 1.53 \text{ s}$

(b)(iii)(B) Improvement: use smaller interval steps for h near the minimum. This would allow more precise identification of the turning point.

Answer: Take more data near minimum T .

(c) Use $T^2 = (4\pi^2/g)(h^2 + C)$.

$$T^2 = (1.53)^2 = 2.341.$$

$$(T^2g)/(4\pi^2) = 0.581.$$

$$C = 0.581 - h^2 = 0.491.$$

Answer: $C \approx 0.064$

Final answers

(a) Measure many oscillations and divide

(b)(ii) $h \approx 0.41 \text{ m}$ and 0.20 m

(b)(iii)(A) $T_{\min} \approx 1.53 \text{ s}$

(b)(iii)(B) More data near minimum improves accuracy

(c) $C \approx 0.064$

Revision tips

- Reaction time errors reduced by timing many oscillations.
- Graphical methods can identify trends and minima.
- Period of physical pendulum depends on distance from pivot.
- Equation: $T^2 = (4\pi^2/g)(h^2 + C)$.
- Plotting T^2 vs h^2 should give straight line with intercept C.